pH and the pH Scale

Answer key with worked solutions

A. Understanding

1. Classify.

> 2.5 acidic; 7.0 neutral; 9.8 basic; 6.9 acidic (just below 7).

2. pH 2 vs pH 5.

> A difference of 3 pH units = 10^3 = **1000 times** greater.

B. Strong acids

3. 0.0010 M HCl.

> $[H_3O^+] = 0.0010$ M; pH = −log(0.0010) = **3.00**

4. 0.25 M HNO_3.

> $[H_3O^+] = 0.25$ M; pH = −log(0.25) = **0.60**

C. Strong bases

5. 0.010 M NaOH.

> $[OH^-] = 0.010$ M; pOH = 2.00; pH = 14.00 − 2.00 = **12.00**

6. 0.0050 M $Ba(OH)_2$.

> Two OH^- per formula unit: $[OH^-] = 2 \times 0.0050\ \text{M} = 0.010\ \text{M}$; pOH = 2.00; pH = **12.00**

D. Concentration from pH

7. $[H_3O^+]$ from pH.

> (a) $10^{-3.20} =$ **6.3×10^{-4} M** (b) $10^{-11.50} =$ **3.2×10^{-12} M**

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8. $[OH^-]$ at pH 5.00.

> pOH = 14.00 − 5.00 = 9.00, so $[OH^-] = 10^{-9.00} =$ **1.0×10^{-9} M**

E. Challenge

9. 100-fold dilution of 0.10 M HCl.

> New concentration = 0.10 M ÷ 100 = 0.0010 M, so pH = −log(0.0010) = **3.00**. The pH rises by **2 units** (from 1.00), one unit for each factor of 10.

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